Lab Experiment: Comparison of Logistic Regression and K Nearest Neighbors (KNN) Classifiers

Aim

To implement Logistic Regression and K Nearest Neighbors (KNN) classifiers on the Breast Cancer Wisconsin (Diagnostic) dataset and compare their performance using standard classification evaluation metrics.

Objectives

1. To preprocess the dataset for classification.
2. To implement Logistic Regression and KNN classifiers using Scikit Learn.
3. To evaluate both models using standard performance metrics.
4. To compare the performance of Logistic Regression and KNN and identify the better classifier.

Dataset

Dataset Name: Breast Cancer Wisconsin (Diagnostic)

Source: UCI Machine Learning Repository

Dataset Link:
https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

Tasks to Perform

1. Download and load the Breast Cancer Wisconsin (Diagnostic) dataset.
2. Perform exploratory data analysis and data preprocessing.
3. Check for missing values and prepare the dataset for classification.
4. Split the dataset into training and testing sets.
5. Train a Logistic Regression classifier.
6. Train a K Nearest Neighbors (KNN) classifier.
7. Evaluate both classifiers using:
    - Accuracy
    - Precision
    - Recall
    - F1 Score
    - Confusion Matrix
8. Compare the performance of the two classifiers using a comparison table.

9. Interpret the results and comment on the better-performing classifier for the given dataset.

Expected Learning Outcomes

Upon successful completion of this experiment, students will be able to:

1. Implement Logistic Regression and KNN classifiers using Scikit Learn.
2. Apply preprocessing techniques to a real world dataset.
3. Evaluate classification models using standard performance metrics.
4. Compare the performance of multiple classifiers and interpret the results.
5. Recommend the most suitable classifier based on experimental findings.